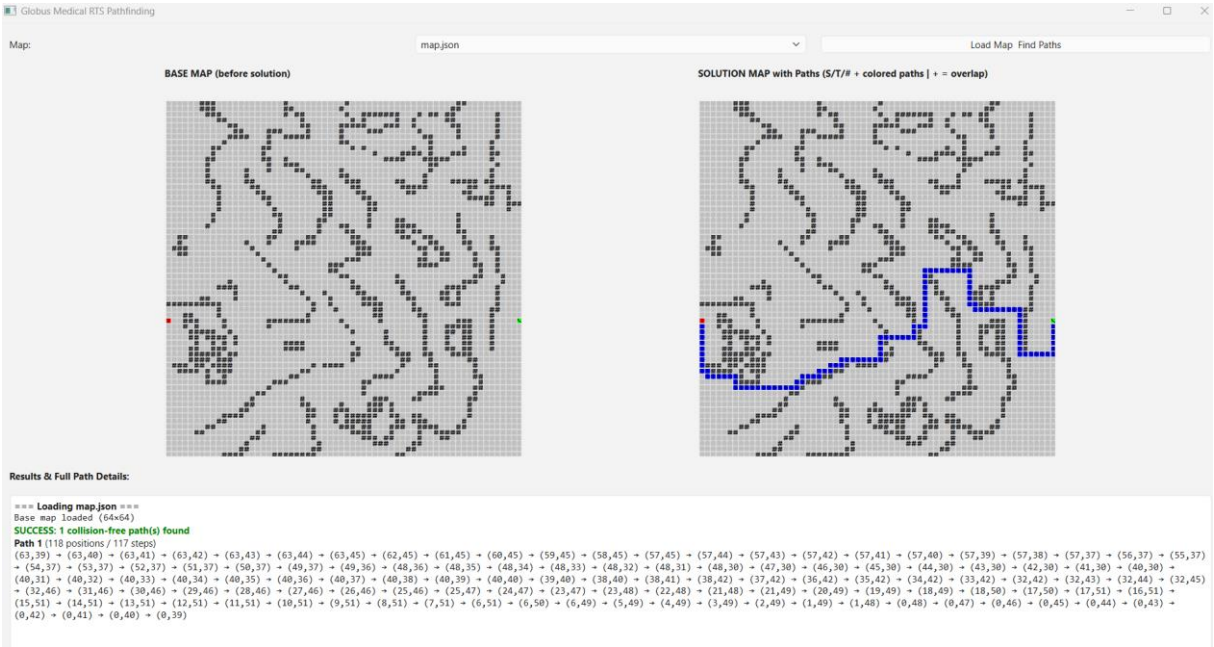
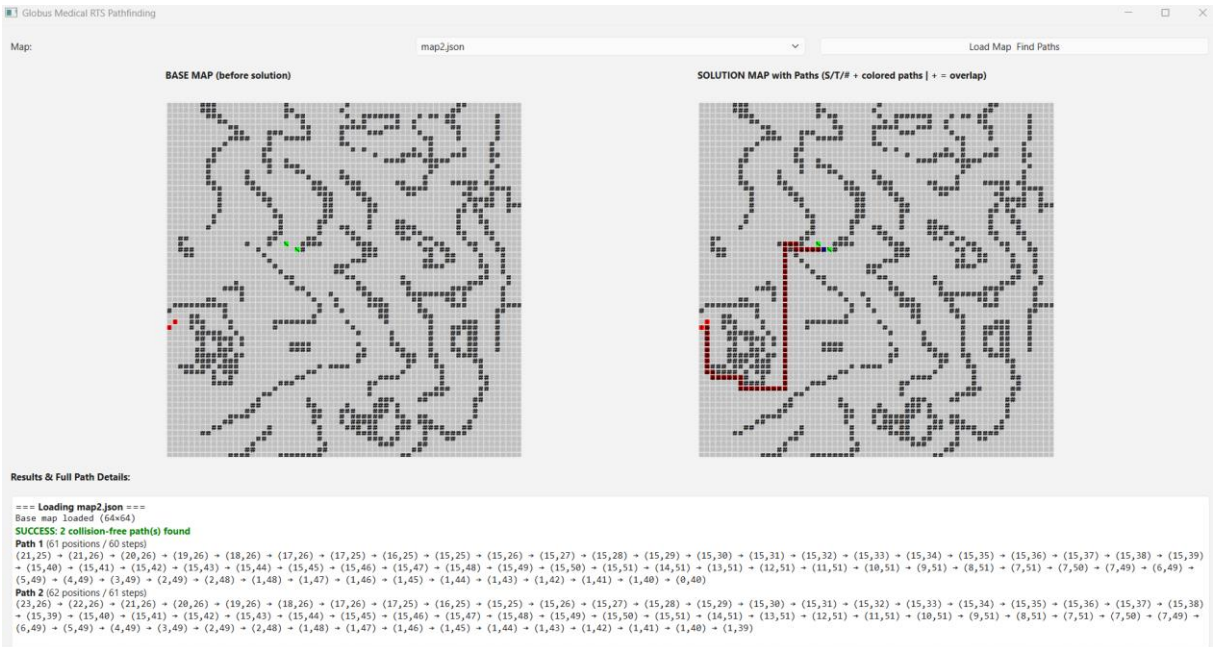
Results

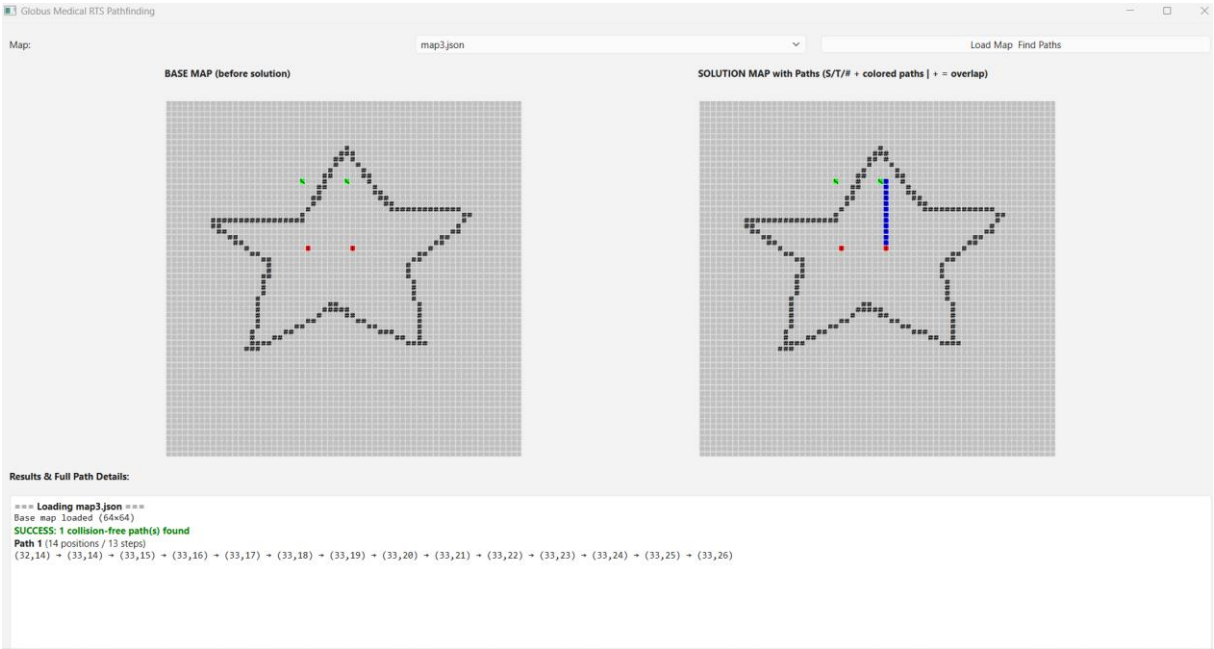
map.json



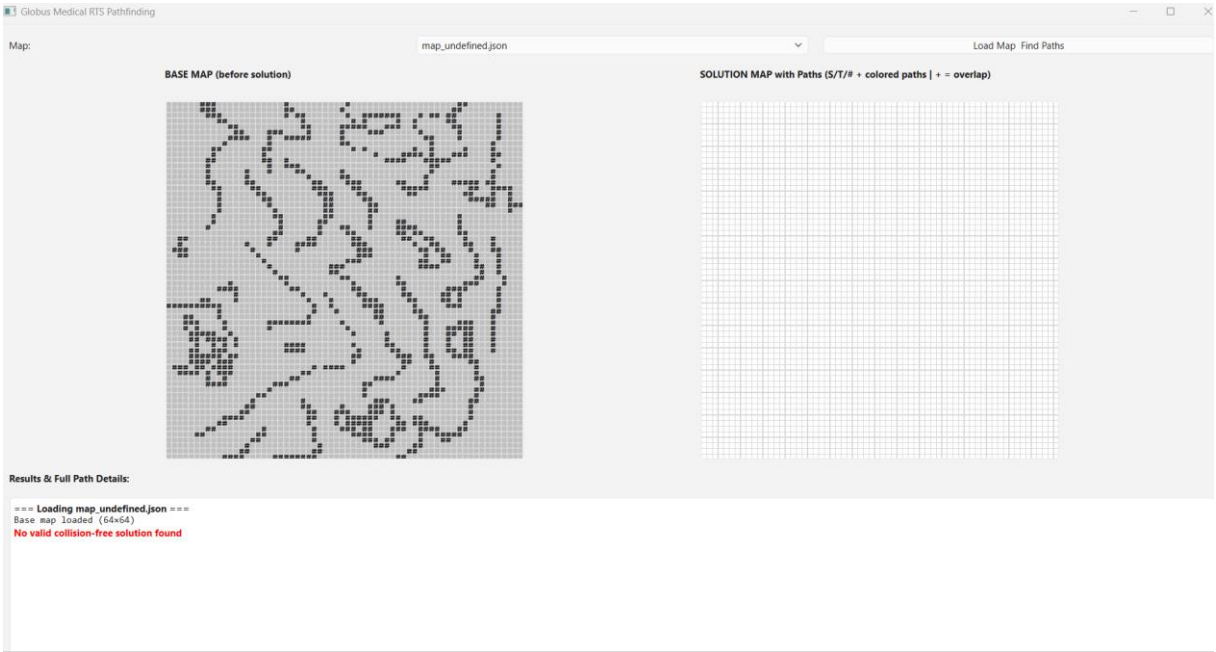
map2.json



map3.json



map_undefined.json



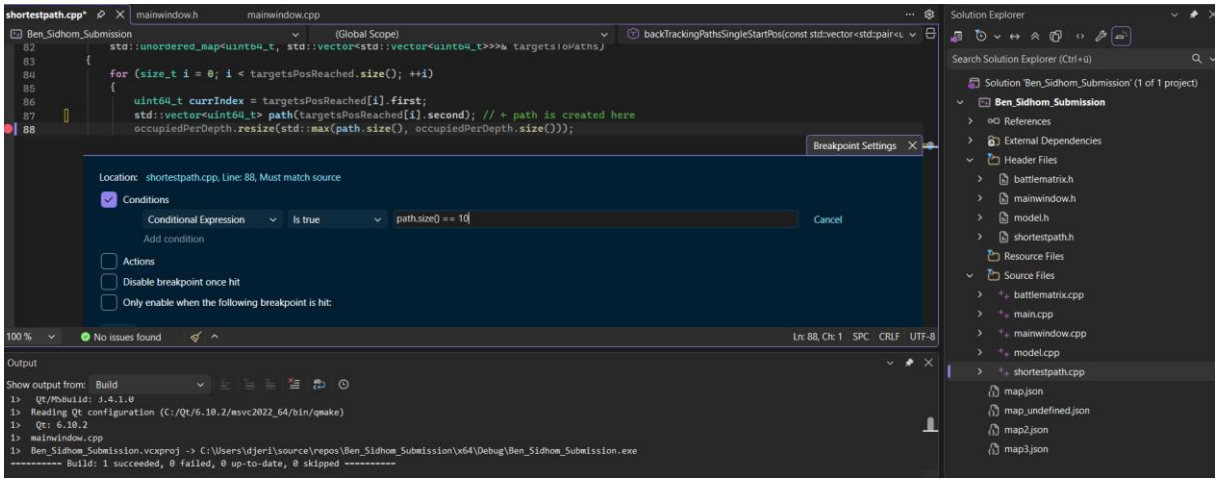
Debugging Tasks

Task 1: Insert a breakpoint when a path is exactly 10 fields long + document local variables

Case 1: *(*I assumed you meant case 2)

Breakpoint when path == 10 fields long (the full path)

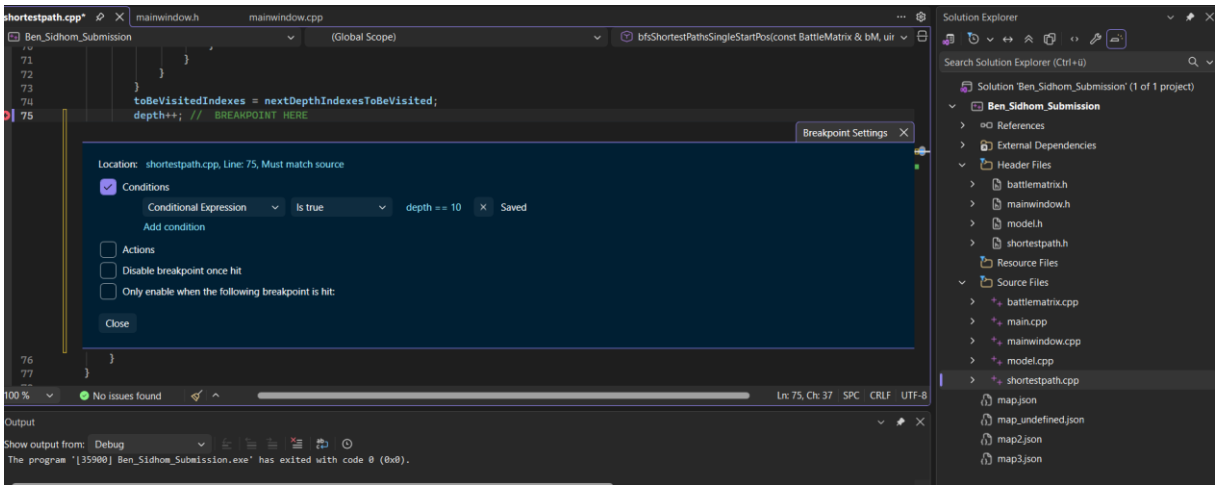
- Location: backTrackingPathsSingleStartPos – after path[path.size()-1-j] = currIndex;
- Condition: path.size() == 10

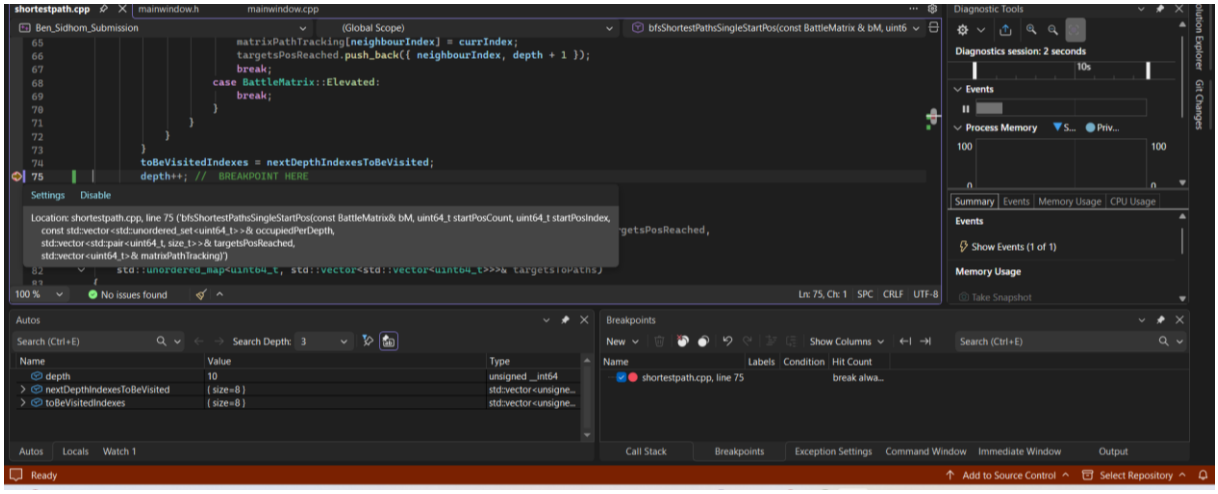


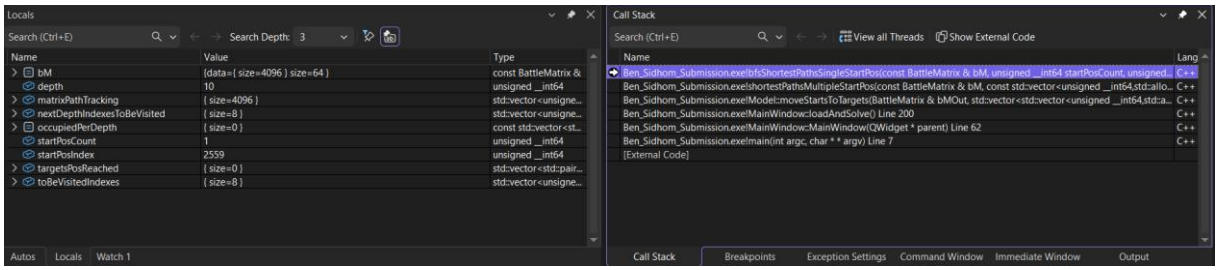
Case 2:

Breakpoint when path reaches 10th step

- Location: shortestpath.cpp inside bfsShortestPathsSingleStartPos
- Line: depth++;
- Condition: depth == 10





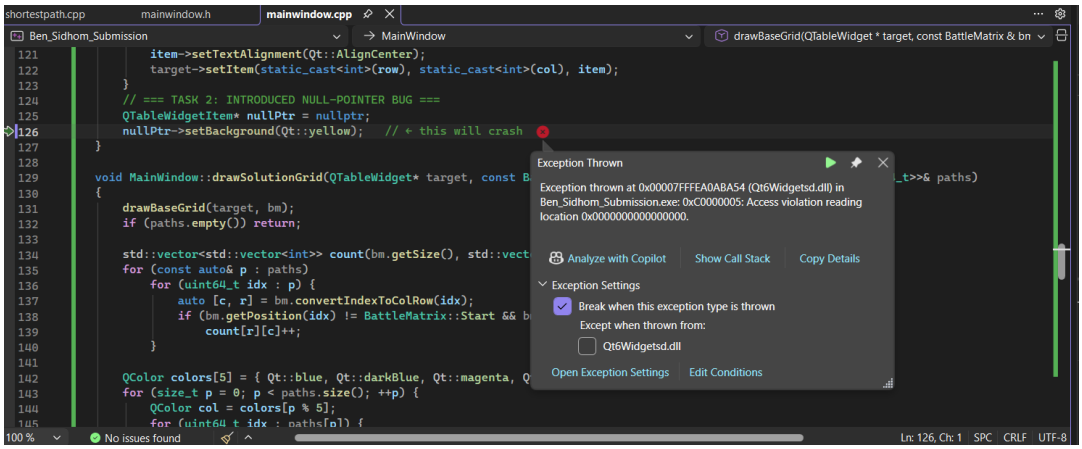


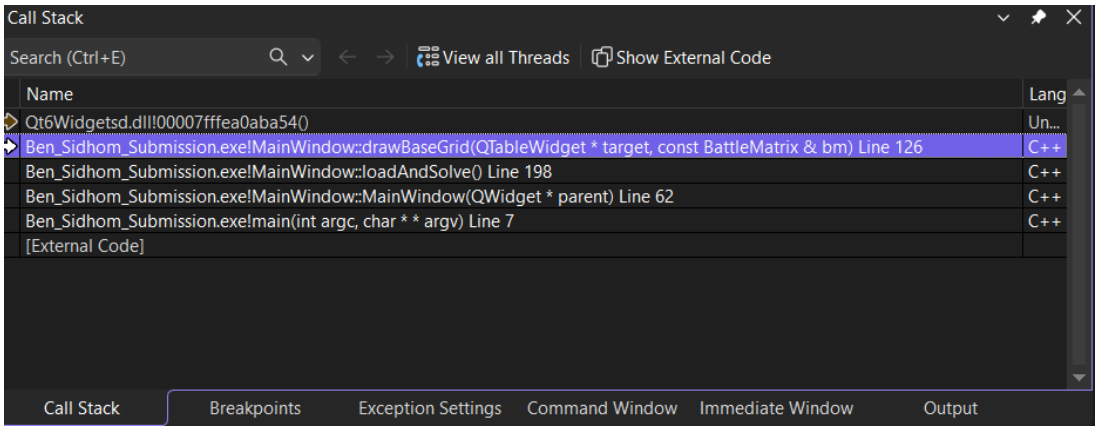
Task 2: Introduce a null-pointer dereferencing bug + show call stack in debugger

Added in mainwindow.cpp inside drawSolutionGrid:

QTableWidgetItem nullPtr = nullptr;*

nullPtr->setBackground(Qt::yellow);



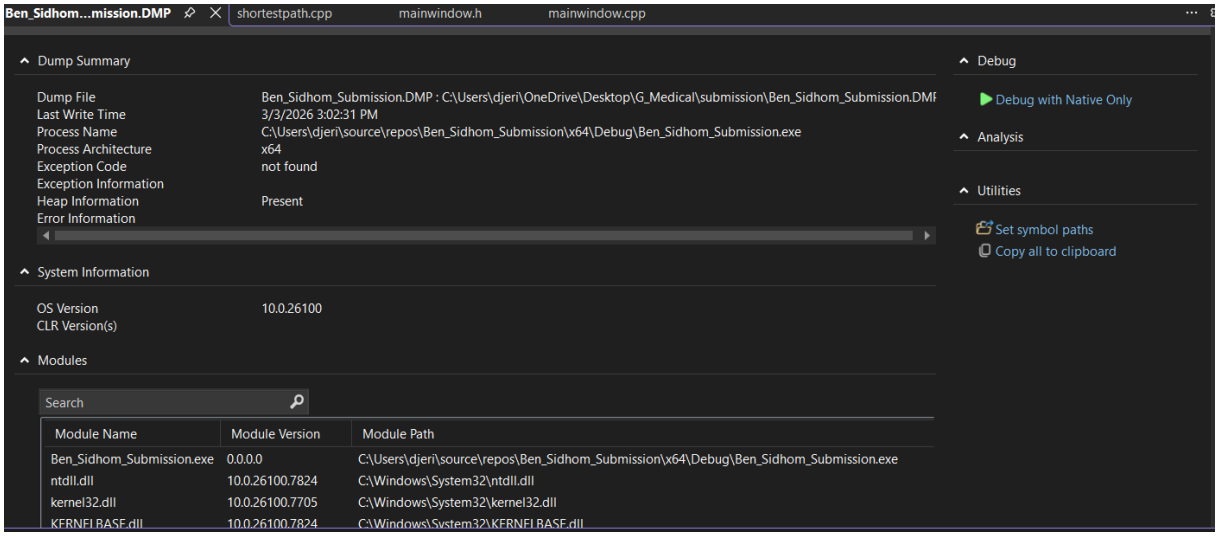


Task 3 : Let it crash and extract call stack from a Windows minidump

Crashed application without debugger

Generated minidump via Task Manager

Opened .dmp file in Visual Studio → confirmed same call stack



Ahmed Ben Sidhom